Hypothesis-Tree Molecular Agents for GuacaMol Amlodipine MPO

Research Proposal

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1 Introduction

GuacaMol's **Amlodipine MPO** asks an agent to propose molecules that score highly under a black-box multi-property objective built around similarity to the antihypertensive drug Amlodipine and a preferred ring count. The formula is withheld from the agent: it only sees SMILES strings and numeric scores. Classical generative baselines often burn tens of thousands of oracle calls; our premise is that a *scientific* agent—one that reads, hypothesizes, experiments, and writes—can climb the same landscape with far fewer evaluations.

We cast discovery as a **hypothesis tree**. Each *node* is a hypothesis experiment that ends in a short research paper (methods, measured matrix, observation, conclusion). Each *edge* is temporal: a child is proposed after its parent is committed, so the tree is an explicit, searchable laboratory notebook rather than a flat list of samples. A single LLM (Qwen) rotates through controller, planner, observer, and reporter roles, using tools to browse memory and submit structured actions. On our Amlodipine campaign the incumbent rose from a seed score of $\mathbf{0.368}$ to $\mathbf{0.905}$ in **83** oracle calls and **48** papers—evidence that autonomous scientific loops can be both sample-efficient and chemically meaningful.

2 Action space

The agent never free-forms tool calls in prose. Every turn returns one JSON envelope {`reason, action, args`}. The edited memory and decision surface we emphasize is:

- `inspect_tree` — paginated gists (title, best molecule, outcome).
- `read` / `search` / `compare` — paper chunks, hybrid retrieval, multi-node facts.
- `deep_read` — dedicated reader model; saves a note (not the raw paper dump).
- `select_parent` / `propose_hypotheses` — pick a line of work; spawn two sibling claims.
- `submit_matrix` — freeze bases×strategies candidates (oracle runs *after* both siblings plan).
- `submit_paper` — commit the experiment paper into the git-backed archive.

Figure 2 shows how often these actions fired on the thinking-on Amlodipine campaign. Memory browsing dominates early; matrix and paper submits track the 48 completed experiments.

3 An agent's trajectory

A concrete path from the live tree:

1. **Level 0 (root).** Seed with Amlodipine itself (score 0.368). Chemically relevant, but not optimal under the ring-aware objective.

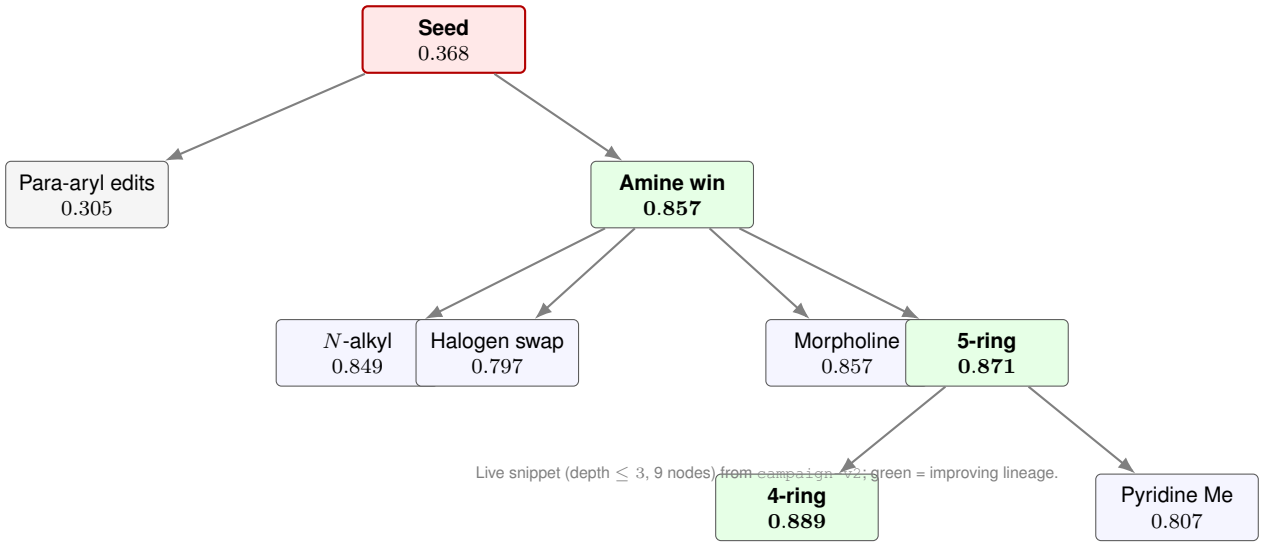


Figure 1: Hypothesis tree from the Amlodipine run. The red root is the seed; green nodes track the rising incumbent through depth 3.

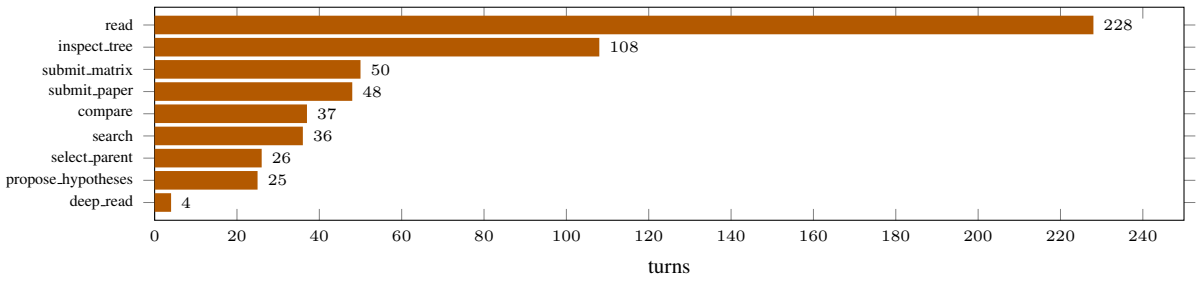


Figure 2: Edited action histogram (thinking-on Amlodipine).

2. **Level 1.** Para-aryl bulk fails (0.305). Replacing the terminal primary amine with a piperazinyl side chain **jumps to** 0.857.
3. **Level 2.** Children fence that win; shrinking the distal diamine toward a five-membered ring lifts the incumbent to **0.871**.
4. **Levels 3–4.** Further ring compression (4-membered, then 3-membered diazacyclopropane) reaches **0.905**.
5. **Deeper levels.** Later papers mostly defend the optimum with near-neighbors and controls.

4 Conclusion: what improved

Starting from Amlodipine (0.368), the agent discovered that the under-scoring axis was largely the **terminal amine / distal ring geometry**, not the pyridine diester core. Keeping that body and installing a compact three-membered diazacyclopropane on the side chain lifted the score to **0.905** after only **83** black-box calls. That is the collaboration hook: **autonomous agents that write real experimental memory can design better molecules at a fraction of the usual oracle budget**, and every step remains inspectable as a paper in a tree.

Open questions

1. **Steering tool use with SFT / RL.** Figure 2 shows a strong bias toward `read`/`inspect_tree`. Can preference optimization or process rewards reshape this histogram—more targeted `search`/`deep_read`, fewer

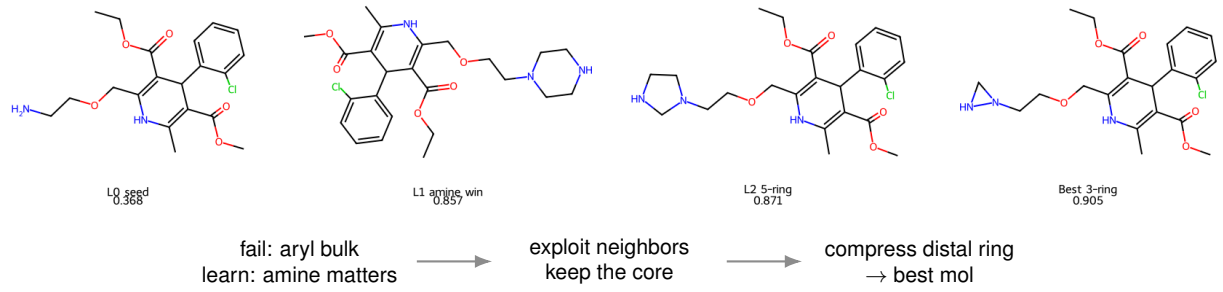


Figure 3: Molecule evolution along the winning lineage (structures, not SMILES): seed → L1 amine win → L2 five-membered distal ring → best three-membered ring ($0.368 \rightarrow 0.857 \rightarrow 0.871 \rightarrow 0.905$), with the learnings that drove each hop.

redundant pages—without hurting chemical progress?

2. **Deeper thinking, richer search.** Beyond document tools, we want better tree-search algorithms and actions such as `think_more` or `try_a_bigger_oracle_call`, so the agent can deliberate longer, spend budget when uncertainty is high, and plan across many papers rather than reacting turn-by-turn.